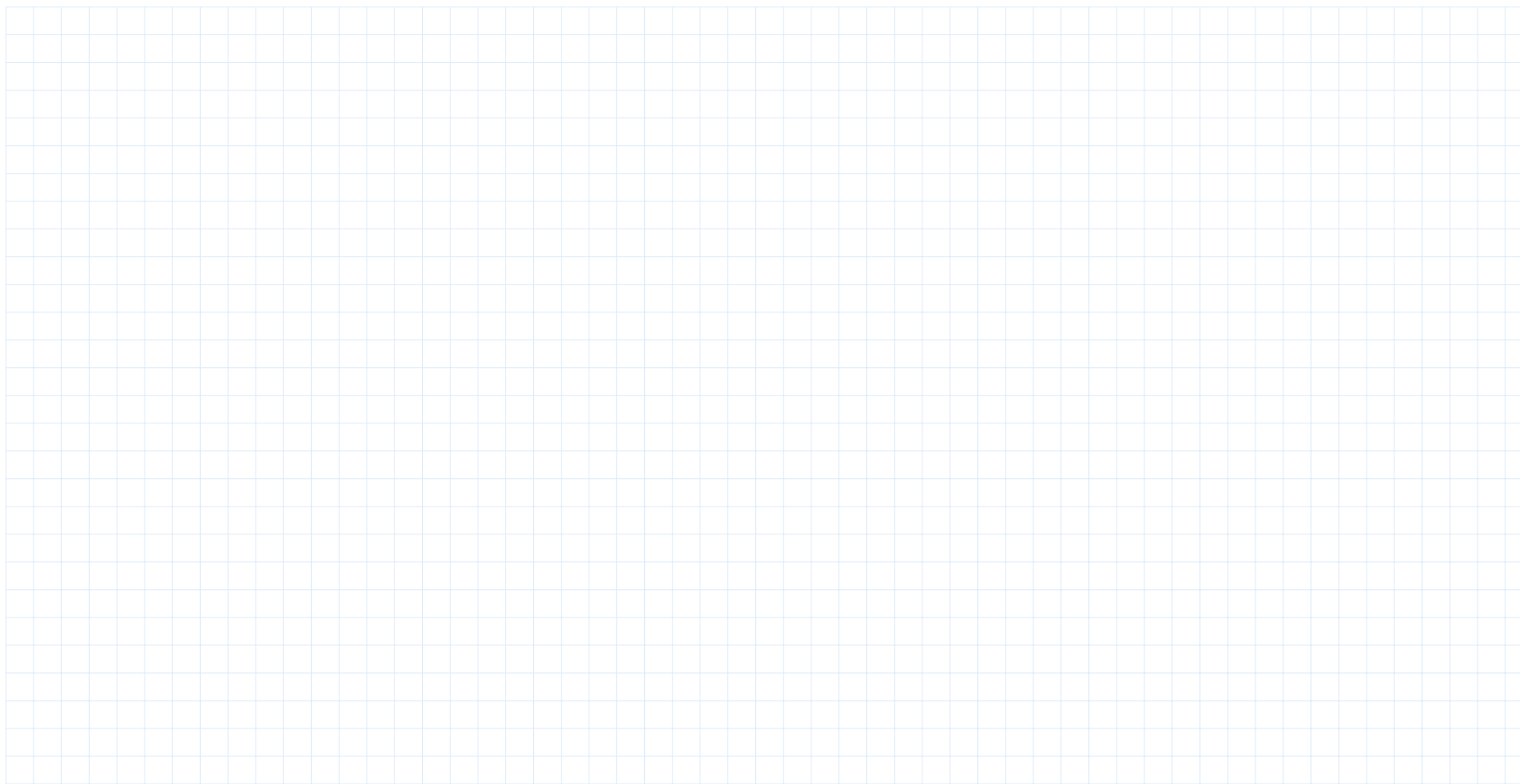


Cosine of a Sum or Difference

The cosine sum and difference formulas are

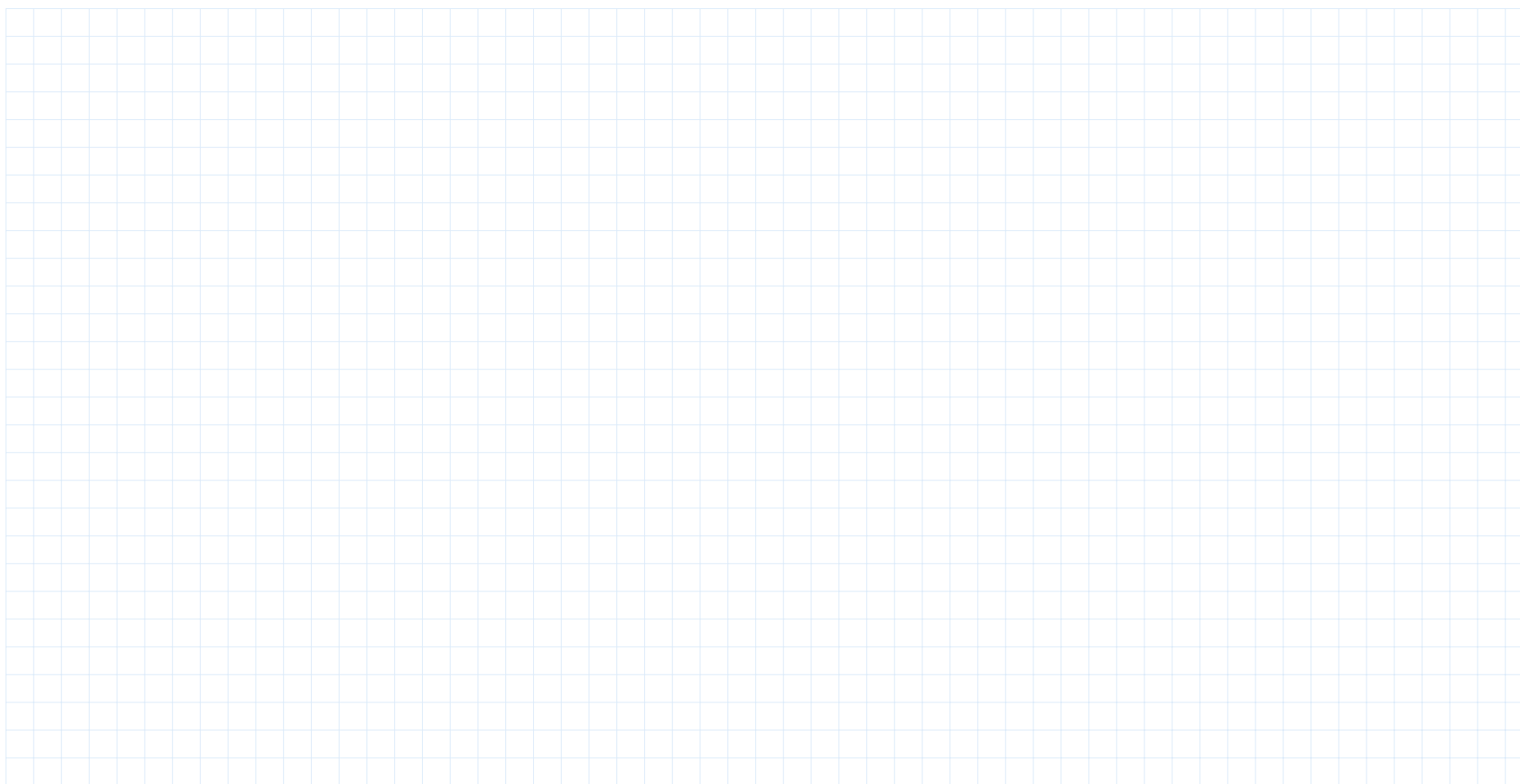
$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta.$$

Use $75^\circ = 45^\circ + 30^\circ$ to find $\cos 75^\circ$ exactly.



$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta.$$

Find $\sin 15^\circ$ exactly using $15^\circ = 45^\circ - 30^\circ$.

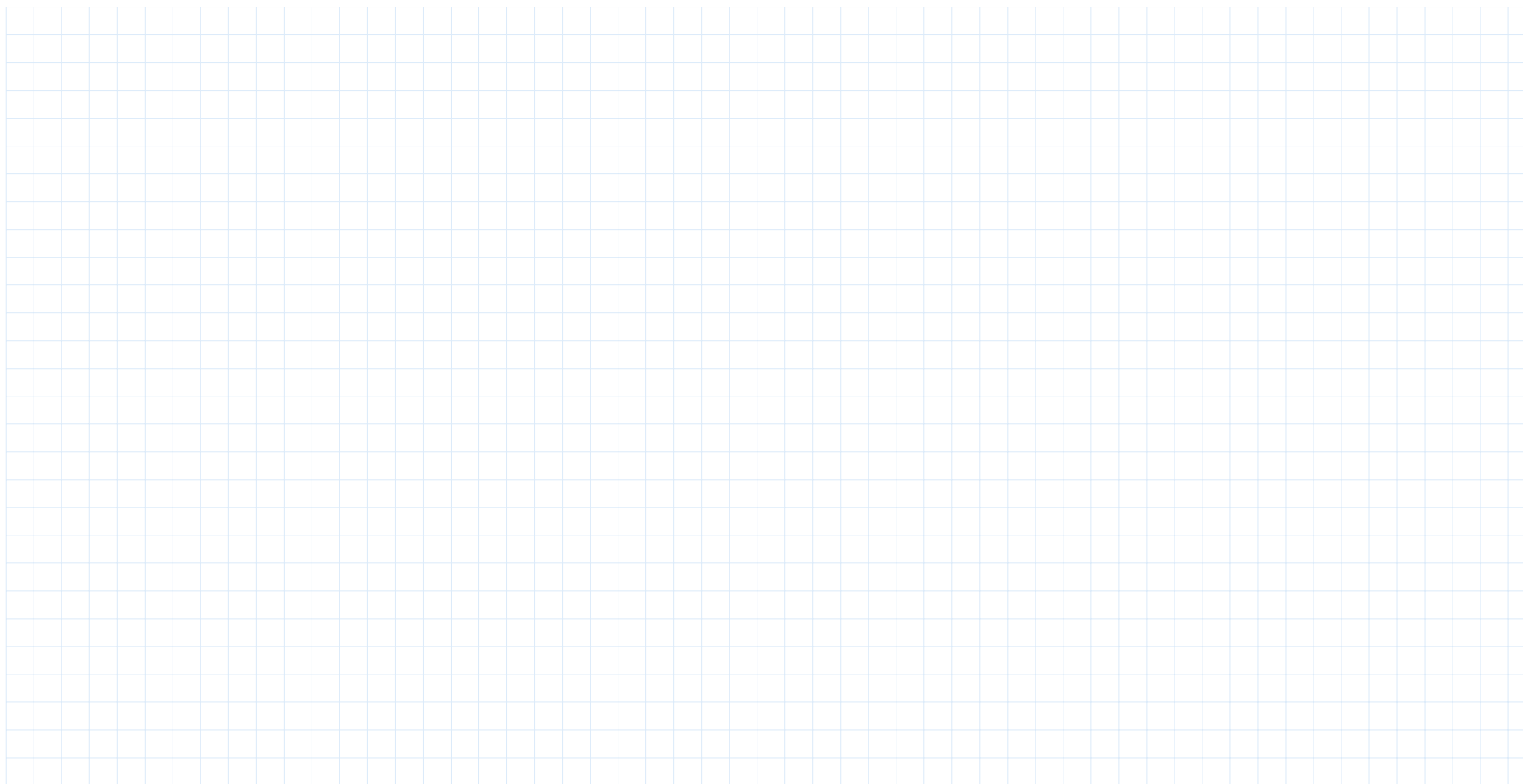


Recognize a Formula Backwards

Rewrite each expression as one trigonometric function, then find its exact value.

(a) $\cos 20^\circ \cos 40^\circ - \sin 20^\circ \sin 40^\circ$

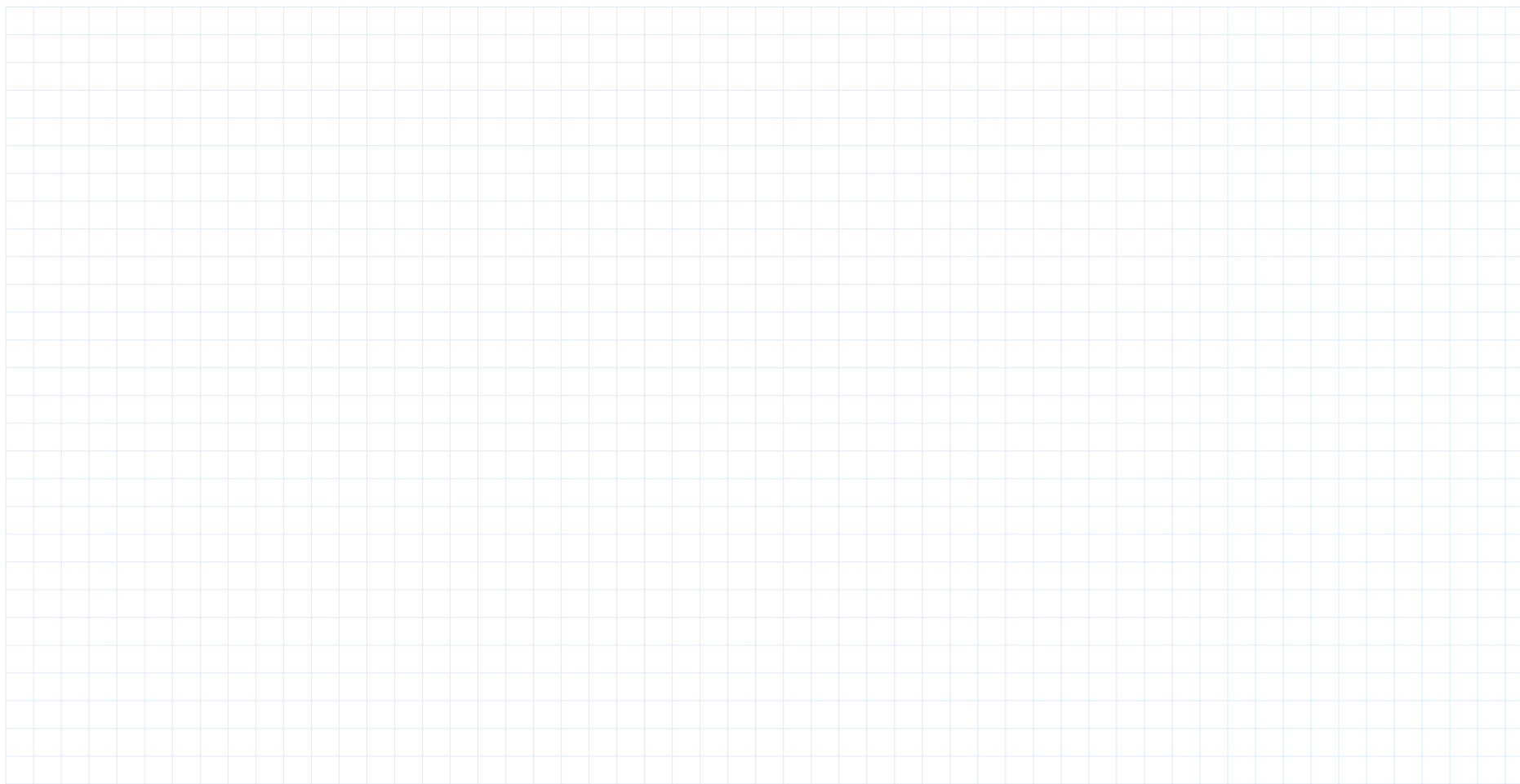
(b) $\sin 20^\circ \cos 70^\circ + \cos 20^\circ \sin 70^\circ$.



Use Quadrant Information

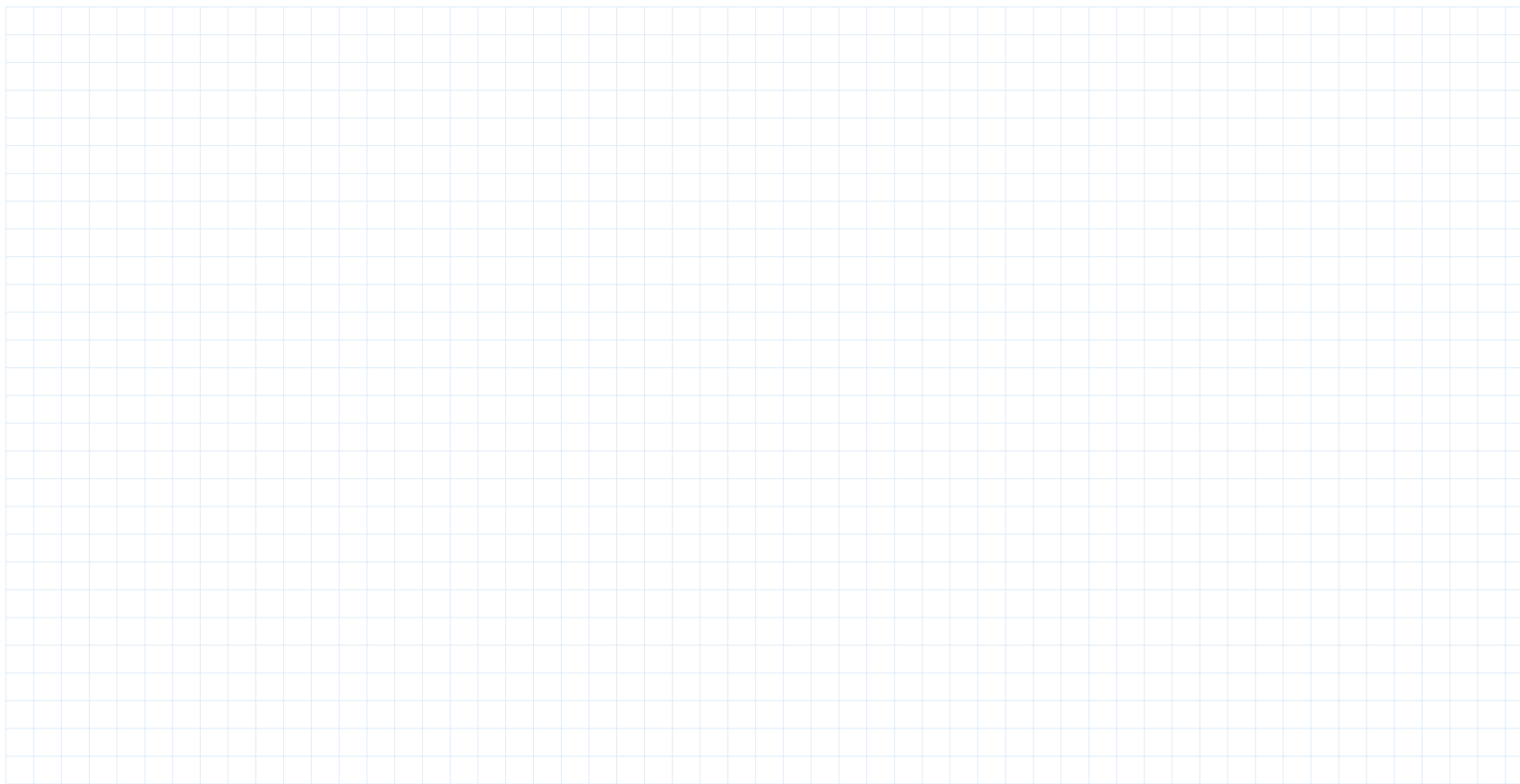
Section 6.5

Suppose $\sin \alpha = \frac{3}{5}$ with $\frac{\pi}{2} < \alpha < \pi$, and $\cos \beta = \frac{5}{13}$ with $\frac{3\pi}{2} < \beta < 2\pi$. Find $\cos(\alpha + \beta)$ and $\sin(\alpha + \beta)$ exactly.



$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta},$$

when the expressions are defined. Find $\tan 75^\circ$ exactly, and simplify the result.

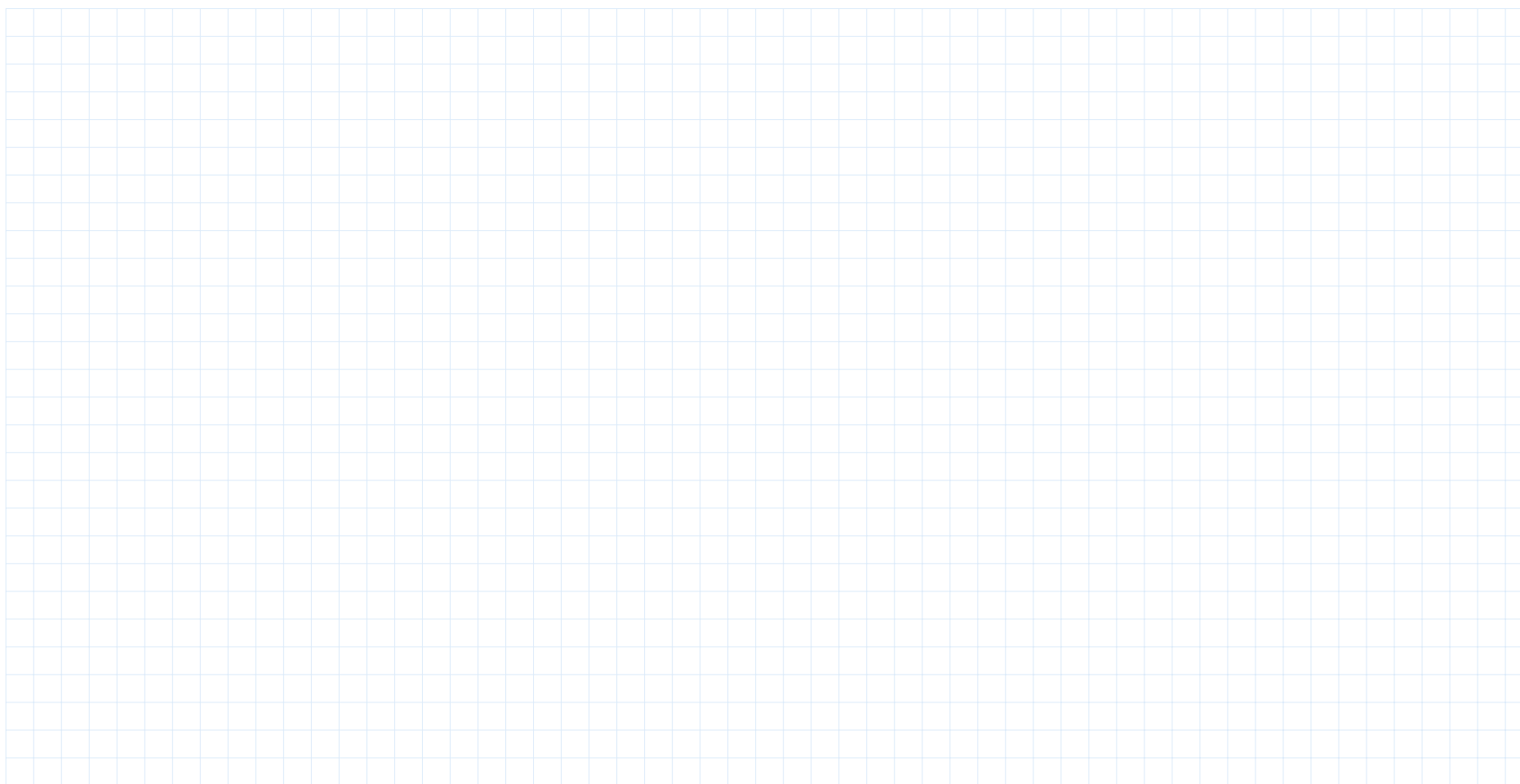


Establish an Angle-Shift Identity

Use a sum formula to establish

$$\sec\left(\frac{\pi}{2} + x\right) = -\csc x.$$

Work on the common domain of both sides.



Combine Inverse Trigonometric Values

Find the exact value without a calculator:

$$\cos\left(\arcsin \frac{3}{5} + \arctan \frac{5}{12}\right).$$

Choose the quadrant of each angle, find the needed ratios, and apply a sum formula.

